

Multi-hazard **Structural Health Monitoring** Bridging Sensing, Data Analytics, and **Digital Twin Systems**

Warren Bridge, RI

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Meets Real Transportation

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THE
UNIVERSITY
OF RHODE ISLAND



THE CHAMPLIN
FOUNDATION



Outline

- Background
- **What Is A Digital Twin?**
- Project Objectives
- Infrastructure Digital Twin Framework
- **Case Study (Warren Bridge, RI)**
- Ongoing Work (Physics-Based Digital Twin)
- Future Work (Hydrodynamic Modeling within Digital Twin Framework)

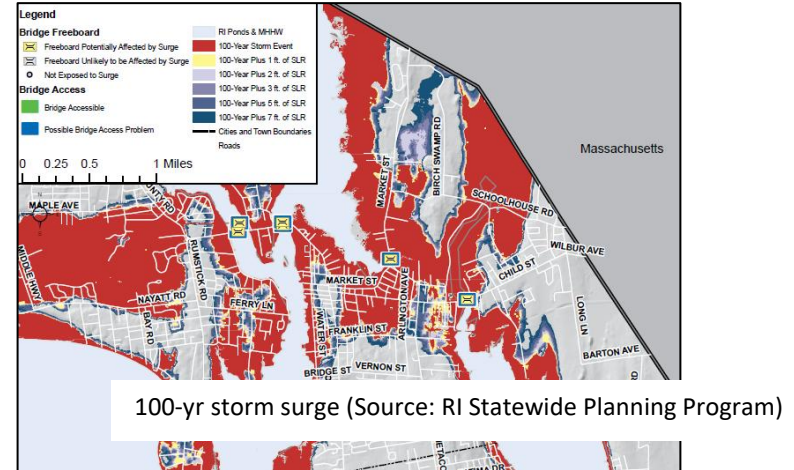
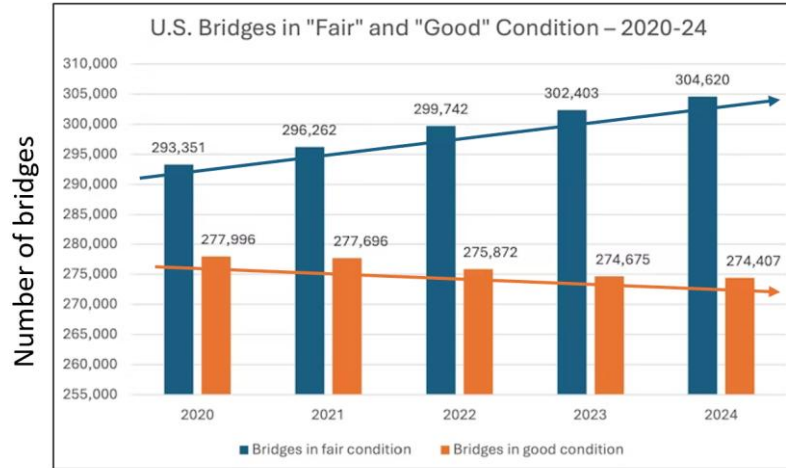
Warren Bridge, RI



Background

- In the U.S., the number of **"fair"** condition bridges has increased, surpassing **"good"** condition bridges. Also, the number of **"good"** condition bridges has decreased in recent years.
- Rhode Island also has over 675 kilometers of coastline, so **many bridges are vulnerable to sea-level rise, storm surge, and scour.**

Source: Federal Highway Administration



- Digital Twin Platform shifts bridge management from reactive inspection and repair toward continuous monitoring, early damage detection, and proactive, risk-informed maintenance.

What Is A Digital Twin?

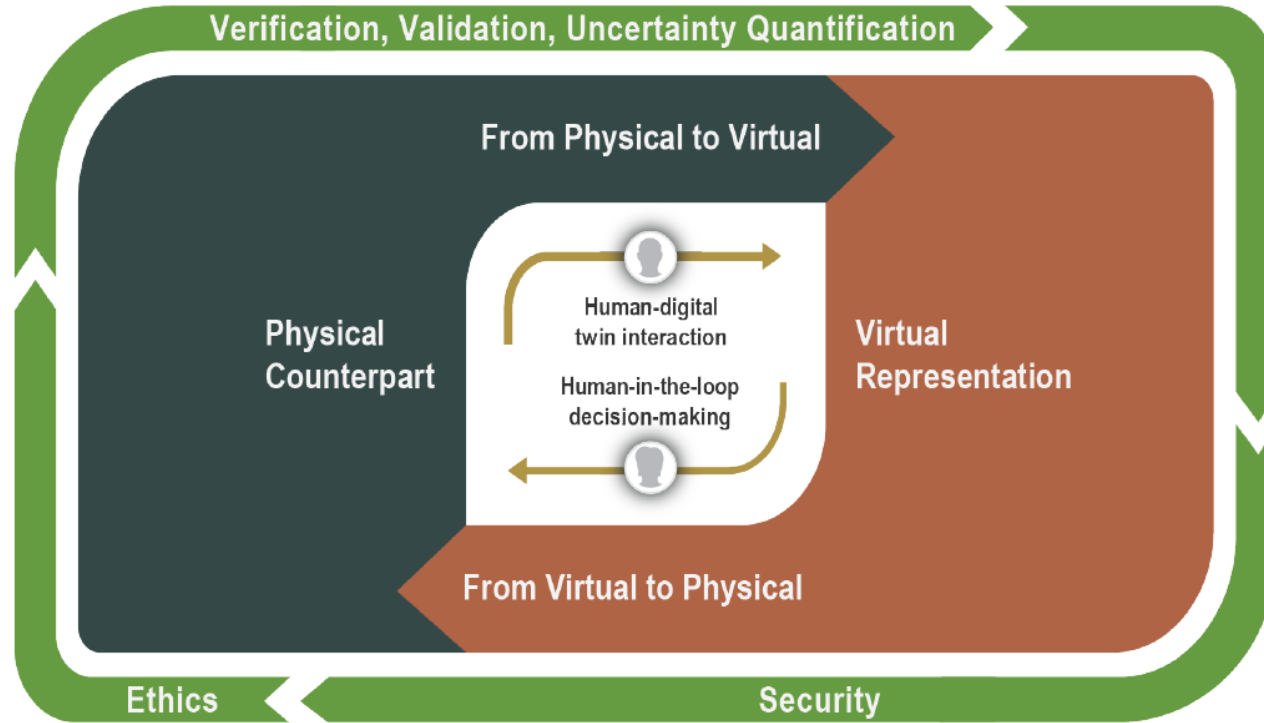
There are several existing definitions for digital twins. Some have been created by researchers, standards committees, consortia, and industry:

- **National Academies (2024):** A digital twin is a set of virtual information constructs that mimics the structure, context, and behavior of a natural, engineered, or social system (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and informs decisions that realize value. **The bidirectional interaction between the virtual and the physical is central to the digital twin.**

What Is A Digital Twin?

There are research

- **National Academies of Sciences, Engineering, and Medicine** structure is dynamic and decision-centric



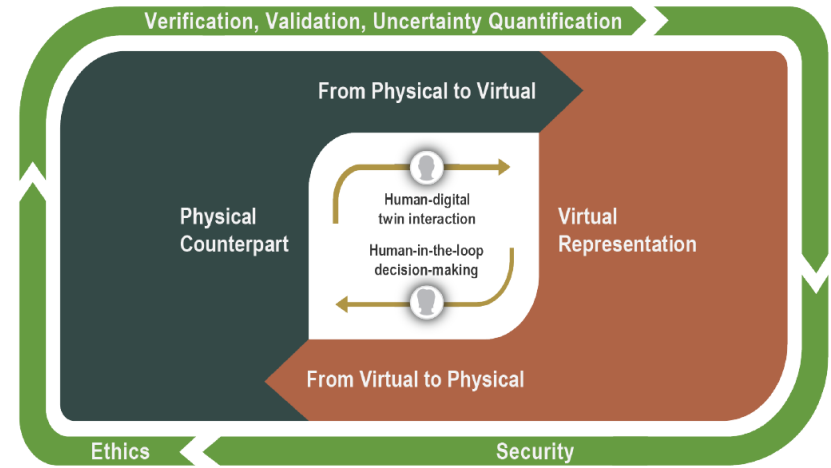
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- A digital twin should be defined at a level of fidelity and resolution that makes it fit for purpose. **Higher fidelity is not automatically better.**
- The appropriate fidelity depends on **(1)** the quantities being predicted, **(2)** computational resources, and **(3)** acceptable cost.



NATIONAL ACADEMIES
Sciences
Engineering
Medicine

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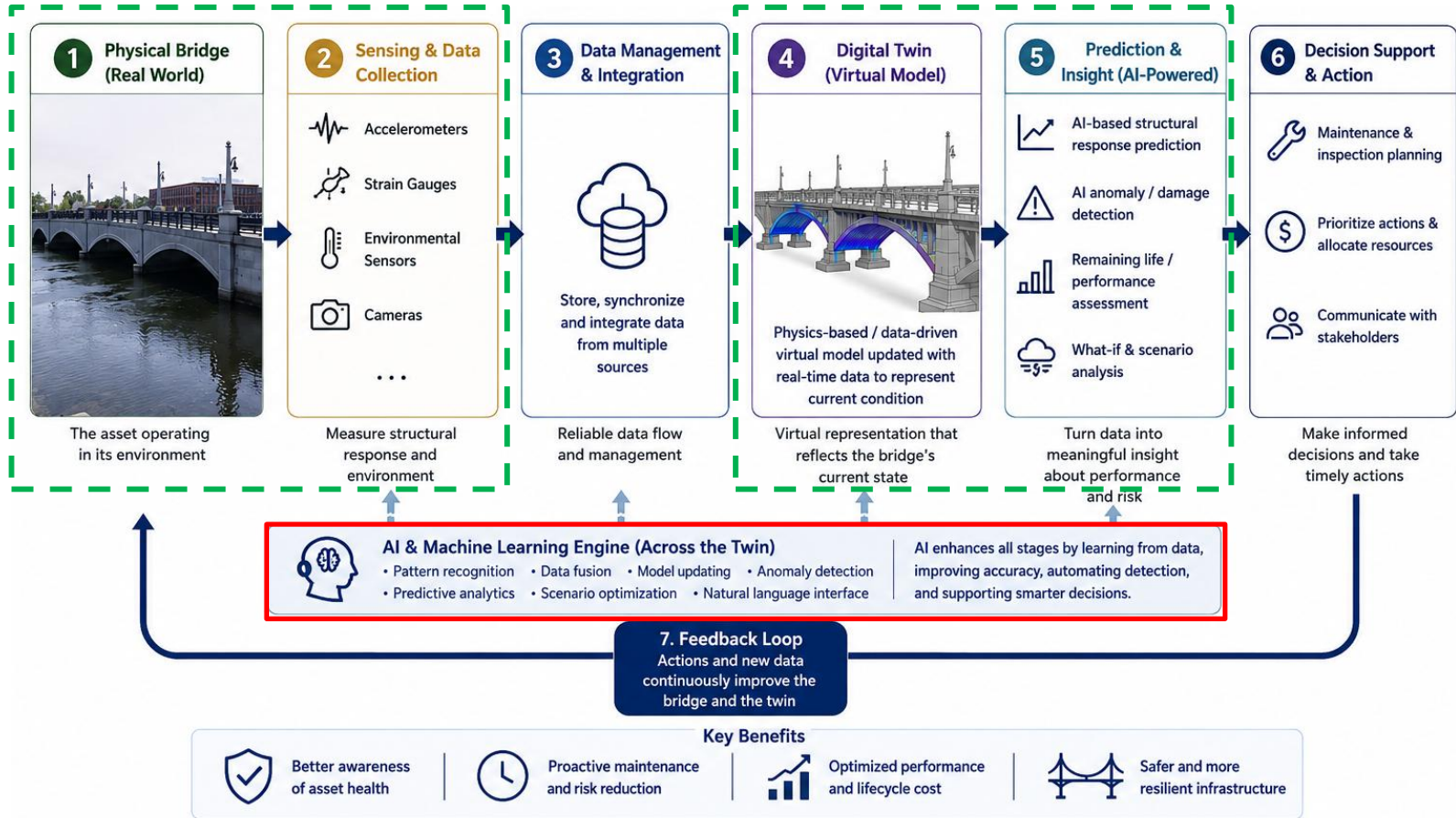
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- **National Institute of Standards and Technology (NIST 2025):** A digital twin is the virtual (i.e., digital) representation of a physical or perceived **real-world entity, concept, or notion.**
- **National Science Foundation (NSF 2026):** A digital twin is a virtual model of real-world things, like buildings, traffic networks, and human organs, **that lets users explore ideas in a low-risk, cost-effective way before taking actions.**
- **Digital Twin Consortium (2020):** A digital twin is an integrated data-driven virtual representation of real-world entities and processes, **with synchronized interaction at a specified frequency and fidelity.**

Project Objectives

- This project initiates an advanced center at the University of Rhode Island (URI) for **Structural Health Monitoring, Data Analytics, and Digital Twin Systems**.



Infrastructure Digital Twin

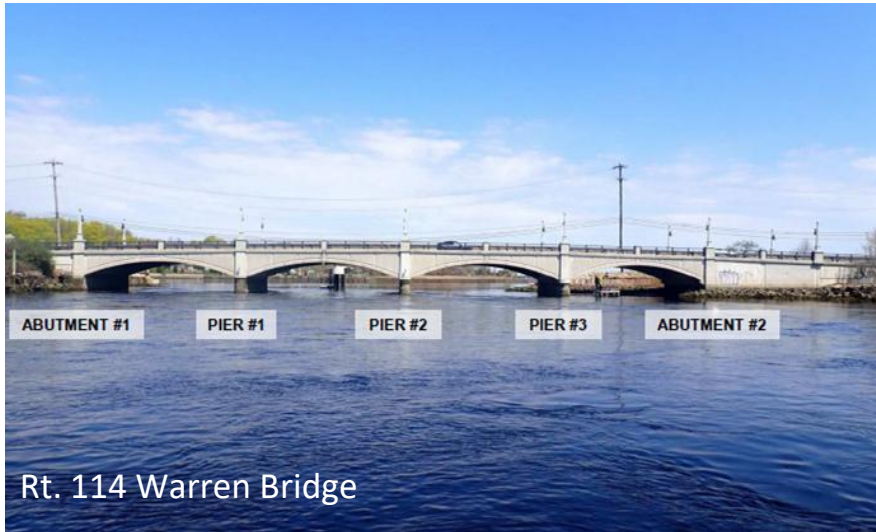


Real-time Data → AI-Powered Virtual Model → Insight → Action → Resilient Infrastructure

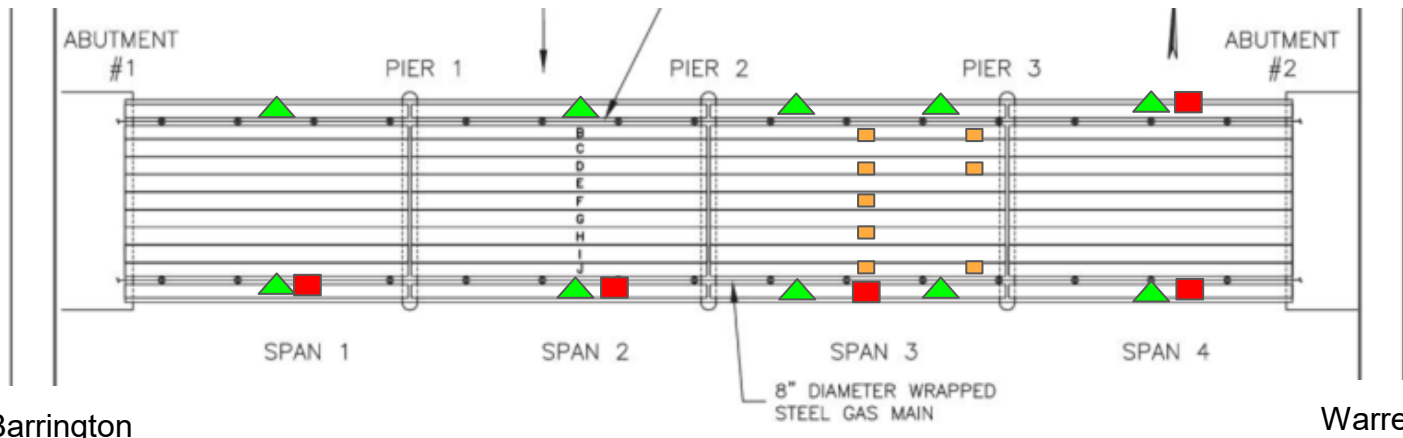
Selected Bridge in RI

Rt. 114 Warren Bridge

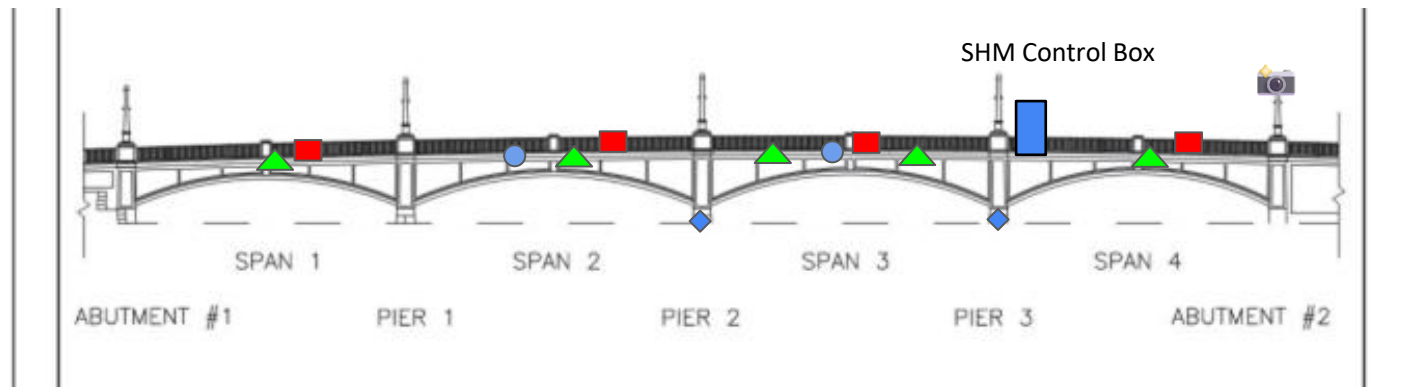
- Prestressed butted concrete box girder; maximum span length: 68 ft; maximum bridge height: 15 ft; number of spans: 4.
- The bridge was built in 1914 -> rehab in 2009.
- Max water depth: 11.1 feet; The bridge is vulnerable to the 100-year storm surge.



Sensor Layout



GNSS (5) (deployed)	
Accelerometer (10)	
Strain gauge (8)	
Sonar depth sensor (2)	
Water level/profile (2)	
Camera (1)	



- GNSS Sensors: Global Navigation Satellite System

Sensor Layout

- **GNSS Sensors:** installed on June 30, 2026
 - Accuracy is about 4mm in the vertical plane and 2mm in the horizontal plane.
 - Sampling rate: 1-hour interval -> **long-term static monitoring.**
 - Solar-based system with satellite connection capability.

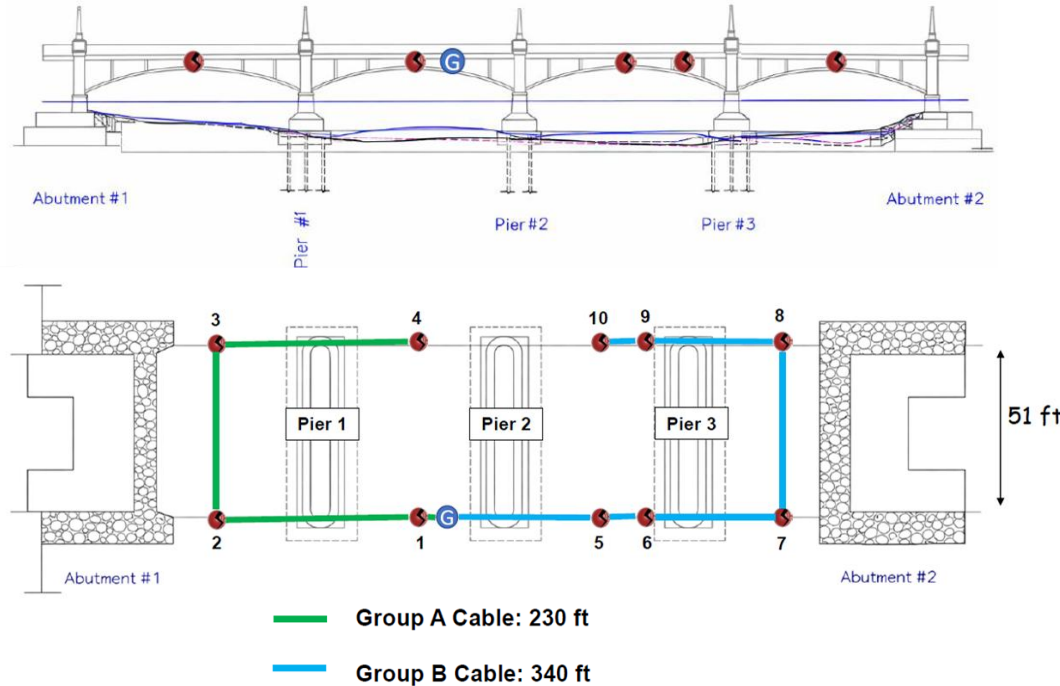


Sensor Layout

- **Accelerometer:** will be installed in Oct. 2026
 - Triaxial velocimeter, ± 120 mm/s
 - Triaxial accelerometer, $\pm 2g$, $\pm 6g$ and $\pm 8g$
 - **Sampling rates from 16 Hz to 266 Hz**
 - **Internet connectivity and power at the gateway**

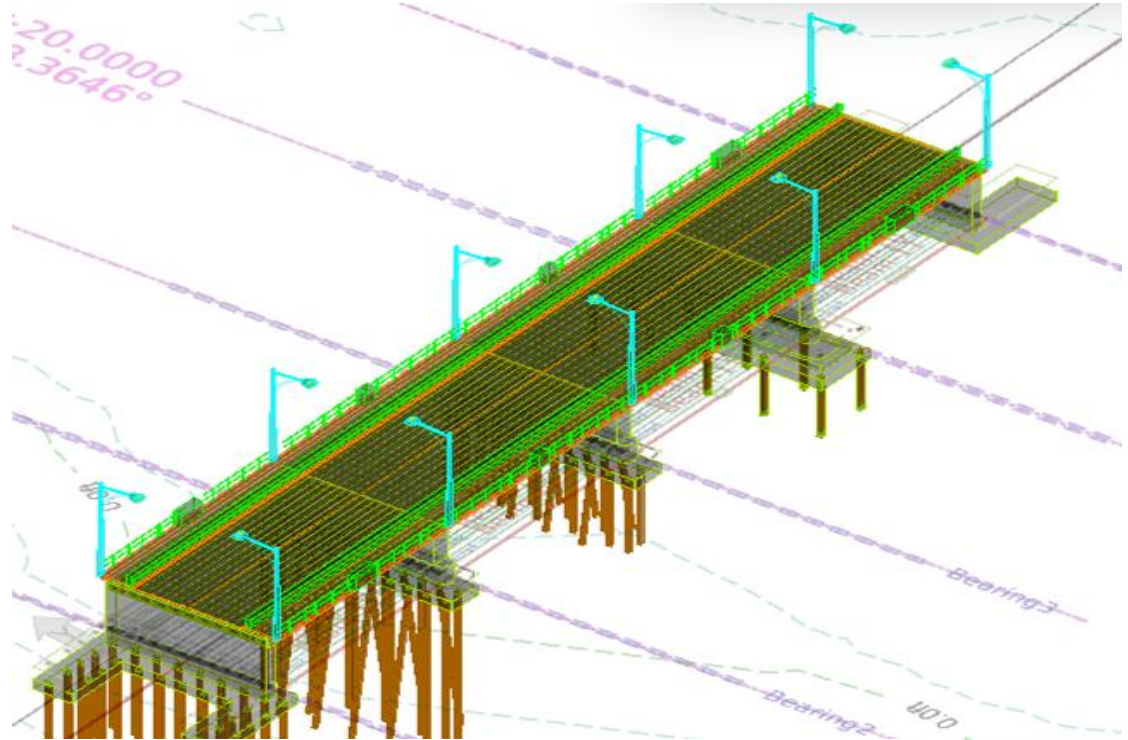


Dimensions: 160 x 110 x 65 mm
Weight: 700 g
Temperature range: -40 to 50°C
Waterproof



Digital Twin Platform (Virtual Bridge Model)

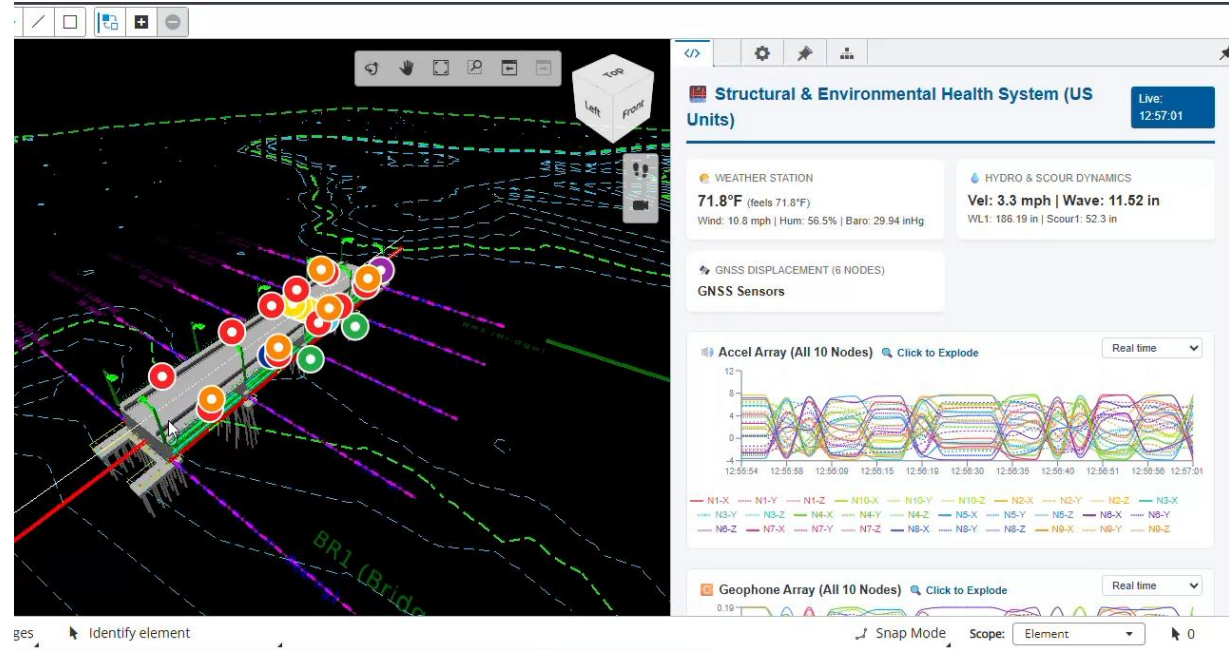
- Developed a detailed model of the Warren Bridge using **Bentley OpenBridge Modeler**.
- The model captures bridge geometry, components, material properties, and surrounding terrain.
- The model provides the foundation for integrating data with the virtual bridge model using **Bentley iTwin** and developing the **Digital Twin Platform**.



Warren Bridge model using Bentley OpenBridge Modeler

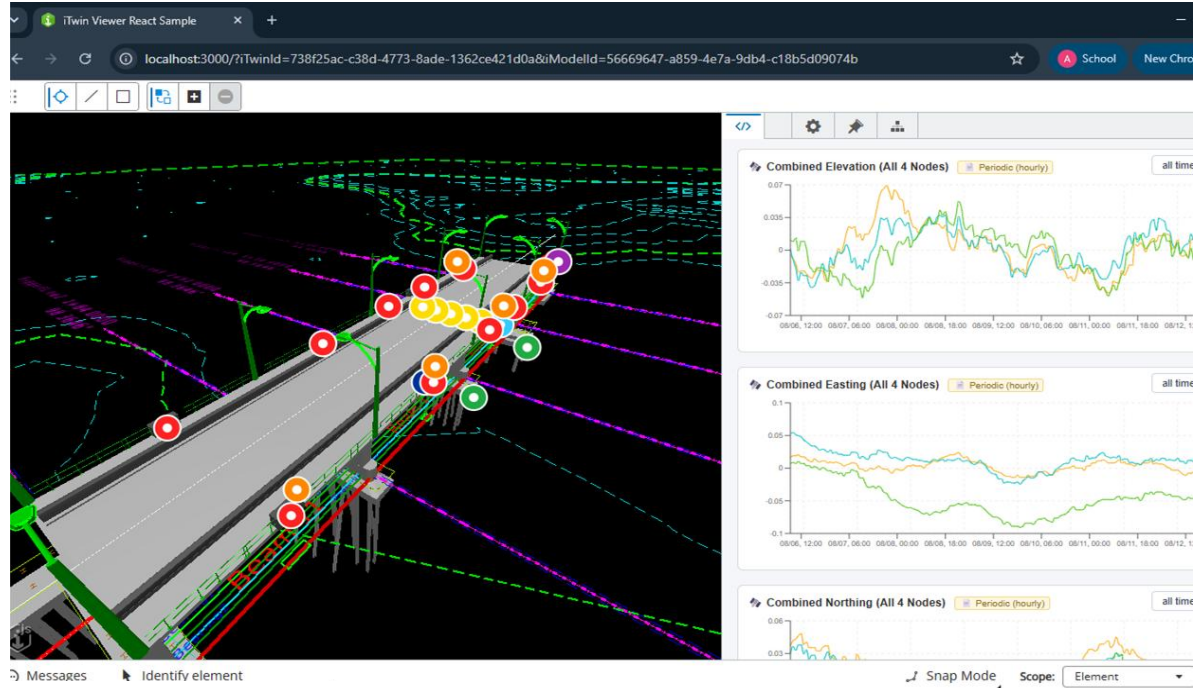
Digital Twin Platform (Ongoing Work)

- Integrated sensor locations and interactive visualization directly within the bridge model.
- **Real-time GNSS data ingestion and visualization as the first live sensor connection.**
- **Next step:** integrate other monitoring data and AI models for condition assessment and anomaly detection.



Warren Bridge model using Bentley iTwin platform

Digital Twin Platform (Ongoing Work)

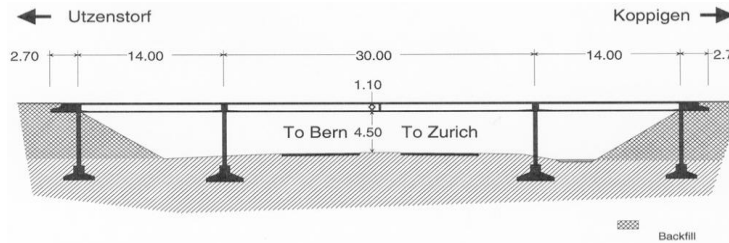


Warren Bridge model using Bentley iTwin platform

Physics-based Digital Twin (Ongoing Work)

OpenSees FE software

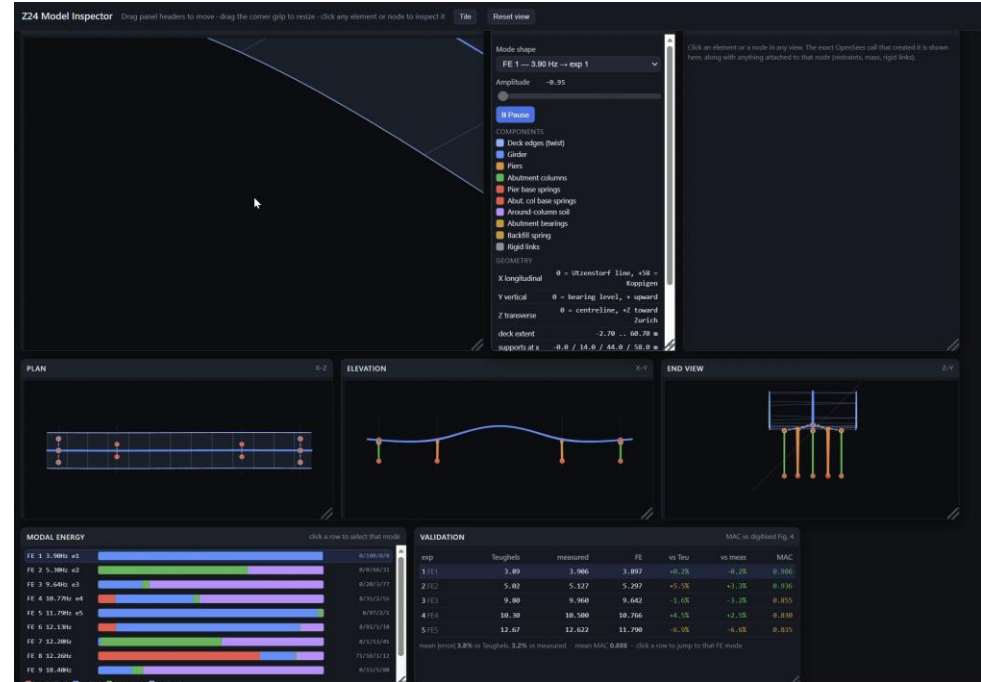
- **Why physics-based?** Connect sensor data to structural behavior to enable **model updating, damage localization, response prediction, and what-if analysis** beyond the conditions directly observed by sensors.
- **Stochastic Subspace Identification (SSI):** obtains natural frequencies, damping ratios, and mode shapes of the structure using acceleration time history data.



Modal Frequencies & MAC — FE Model vs. SSI-Identified Acceleration Data

Mode	Measured (Hz)	FE model (Hz)	Difference	MAC	Mean MAC
1	3.91	3.89	-0.6%	0.987	0.908
2	5.13	5.08	-1.0%	0.968	
3	9.98	9.76	-2.2%	0.911	
4	10.53	10.79	+2.5%	0.847	
5	12.62	11.80	-6.5%	0.826	

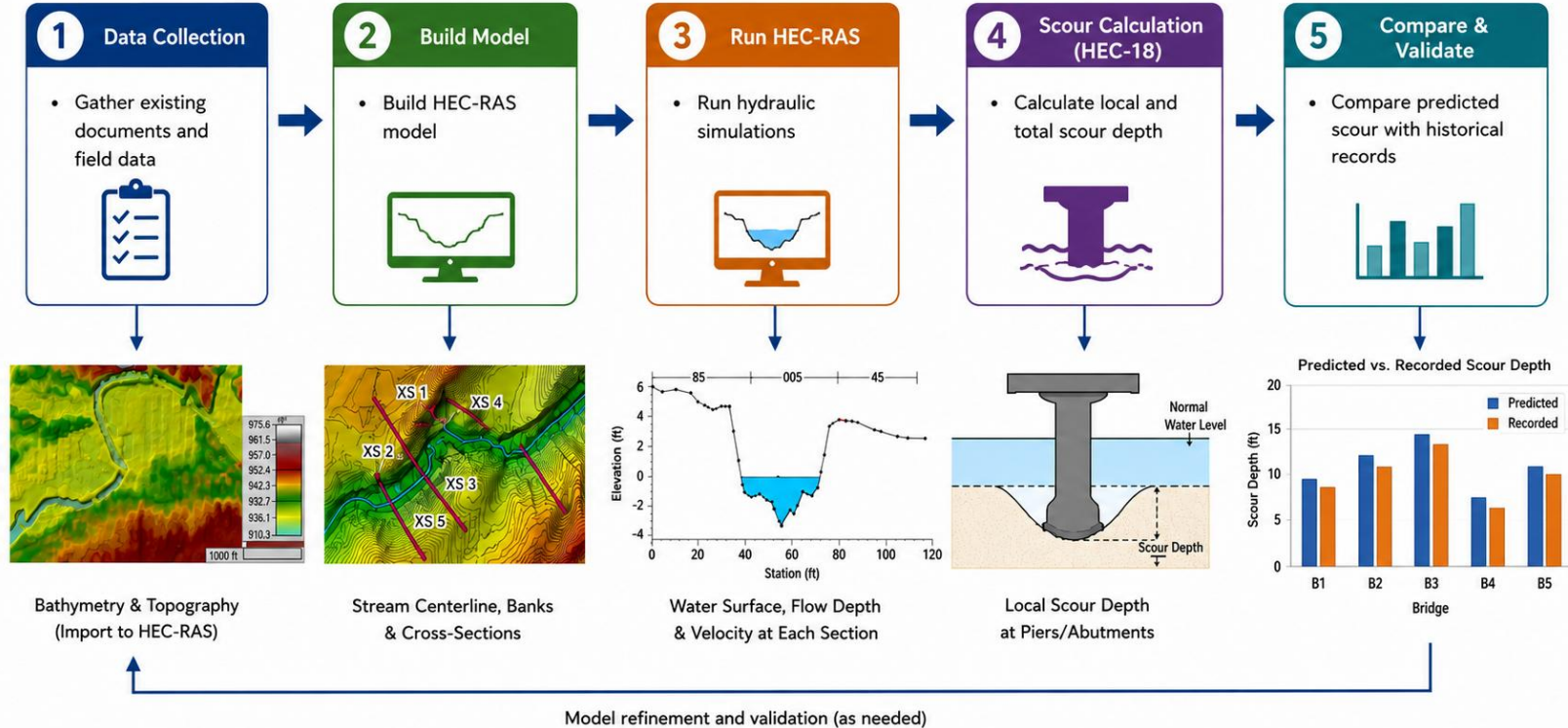
Z24 Bridge benchmark dataset (located in Switzerland)



- We are currently validating the SSI method using the Z24 Bridge benchmark dataset -> next step is to implement it for the Warren Bridge to support physics-based digital twin development.

Physics-based Digital Twin (Future Work)

HEC-RAS Workflow for Bridge Scour Prediction



THANK YOU!
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Questions?

THINK BIG  WE DOSM

